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A Novel Self-Sustaining Process in Plane Couette Flow

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Motivation

Why Study Self-Sustaining Turbulence?

- Wall-bounded turbulence is ubiquitous
- Turbulence is not fully chaotic: coherent structures persist within it
- A well-established self-sustaining process (SSP) has explained wall-bounded turbulence dynamics



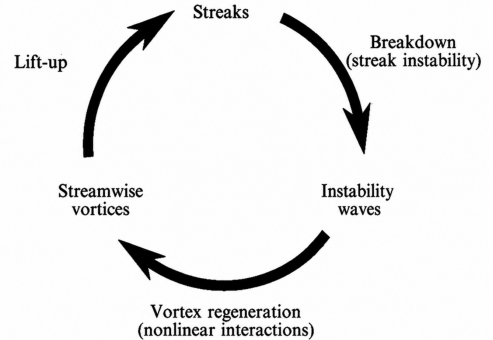
Coherent streak-like structures on a beach

This thesis tests a new process to sustain turbulence.

The Classical Self-Sustaining Process

Hamilton, Kim & Waleffe (1995) [1]

- Closed, three-step loop
- Every structure in the cycle is streamwise-elongated



Schematic of the classical SSP (Adapted from [1]).

The Quasilinear Approximation

- Splits the flow into a low-wavenumber (mean) field and high-wavenumber fluctuations [2]
- Neglects nonlinear scattering between fluctuations in the high-wavenumber equation
- QL_x, retains the **streamwise mean** as the low-wavenumber field
- Reproduces DNS turbulence statistics almost perfectly [3; 4]
- Confirms the classical picture

$$\frac{\partial \mathbf{U}_l}{\partial t} + \mathcal{P}_l[(\mathbf{U}_l \cdot \nabla) \mathbf{U}_l] = -\frac{1}{\rho} \nabla P_l + \nu \nabla^2 \mathbf{U}_l - \mathcal{P}_l[(\mathbf{u}_h \cdot \nabla) \mathbf{u}_h]$$

$$\frac{\partial \mathbf{u}_h}{\partial t} + \mathcal{P}_h[(\mathbf{u}_h \cdot \nabla) \mathbf{U}_l] + \mathcal{P}_h[(\mathbf{U}_l \cdot \nabla) \mathbf{u}_h] = -\frac{1}{\rho} \nabla p_h + \nu \nabla^2 \mathbf{u}_h,$$

Quasilinear Approximation

Is the streamwise pathway the only way for wall turbulence to self-sustain?

Objective & Approach

Introducing QL_z

- QL_z — a quasilinear approximation built on the **spanwise** mean
- QL_z actively suppresses the classical route
- **Aim:** uncover and characterise this new, spanwise-driven self-sustaining process

Methodology

- Solver: **DIABLO** [5]
- Flow configuration: plane Couette flow
- Minimal flow unit required to sustain large-scale motions
- Three configurations simulated:
 - DNS
 - QL_x
 - QL_z

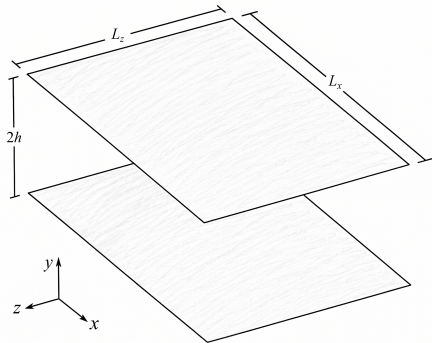
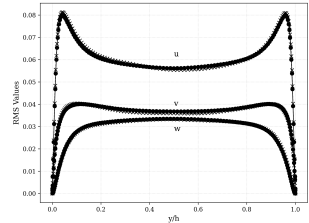
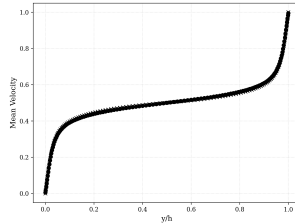
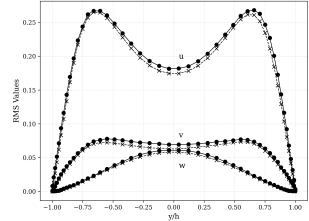
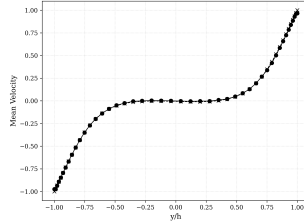


Diagram of flow geometry with coordinate conventions used in the present study

Validation

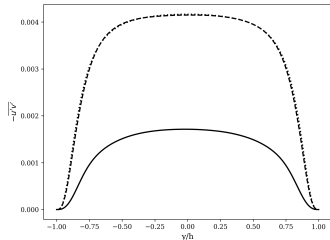
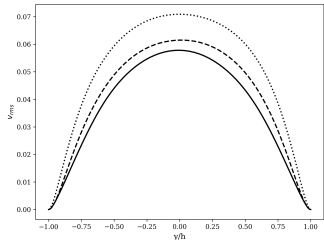
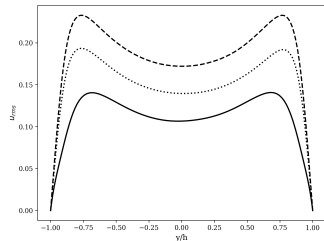
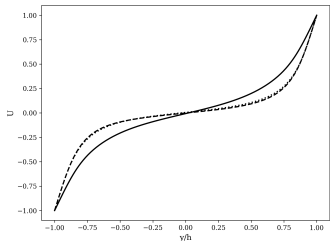
- Benchmarked against two DNS databases [6; 7]
- Compared Mean velocity and RMS fluctuation

Database	$\epsilon_{L_2}(U)$	$\epsilon_{L_2}(u_{rms})$	$\epsilon_{L_2}(v_{rms})$	$\epsilon_{L_2}(w_{rms})$
Kawahara & Kida ($Re_w = 400$)	1.63%	3.38%	5.76%	6.83%
Kawamura Lab ($Re_w = 12800$)	0.97%	1.56%	1.87%	2.28%



QL_z Turbulence Statistics

- $Re_w = 1000$ for DNS and QL_x, and $Re_w = 1428$ for QL_z
- $Re_\tau \approx 66, 65$ and 63 for DNS, QL_x and QL_z respectively
- QL_z still sustains turbulence
- Reduced Reynolds stress and mean shear relative to DNS and QL_x

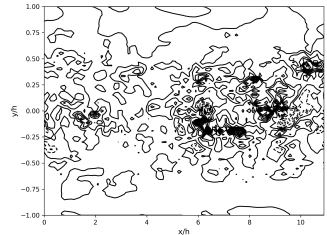
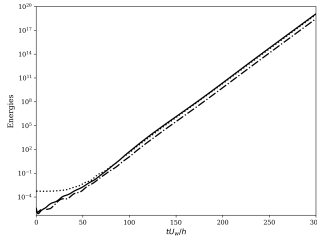
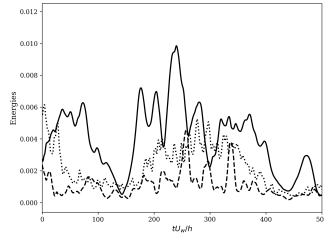


DNS vs. QL_x vs. QL_z: mean velocity, u_{rms} , v_{rms} , Reynolds stress

What Is Driving It?

Base-Flow Stability

- Froze the spanwise-mean mode as a base flow and perturbed it
- The spanwise-mean mode is **linearly unstable** — its breakdown excites the high-wavenumber fluctuations



QL₂ energy evolution; exponential growth under a frozen base flow (streak energy (—————), wavy spanwise-mean mode energy (— — —) and the vortex energy (· · · · ·)); leading eigenmode

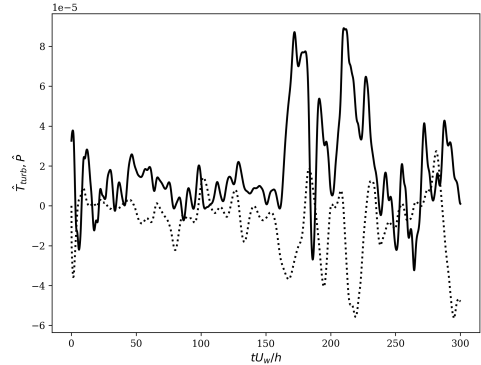
Spectral Energy Balance

$$\frac{\partial \hat{E}(\mathbf{k}, y)}{\partial t} = \hat{P}(\mathbf{k}, y) - \hat{\varepsilon}(\mathbf{k}, y) + \hat{T}_{turb}(\mathbf{k}, y) + \hat{T}_p(\mathbf{k}, y) + \hat{T}_\nu(\mathbf{k}, y)$$

What Energises the Base Flow?

The Active Triads

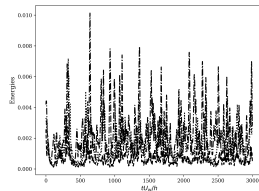
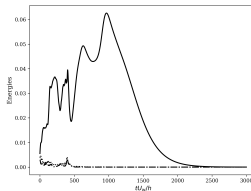
- Spanwise-mean mode $(\alpha, 0)$ energised by turbulent transfer
- Transport is generated by the interaction of the **oblique modes** $(\alpha, \pm\beta)$ with the **streaks** $(0, \beta)$ and higher harmonics



Time evolution of $\hat{T}_{\text{turb}}(\alpha, y, 0)$ component (—), and $\hat{P}(\alpha, y, 0)$ component (· · · ·) energising $(\alpha, 0)$ energy mode

Isolating the Active Triads

- Selectively zeroed specific Fourier modes during the simulation
- Removing the spanwise-mean mode $(\alpha, 0) \rightarrow$ turbulence collapses
- Removing the oblique modes $(\alpha, \pm\beta) \rightarrow$ turbulence also collapses — both are **mandatory**
- Removing the classical streamwise streaks $(0, \beta) \rightarrow$ turbulence **survives** — dynamically **secondary**

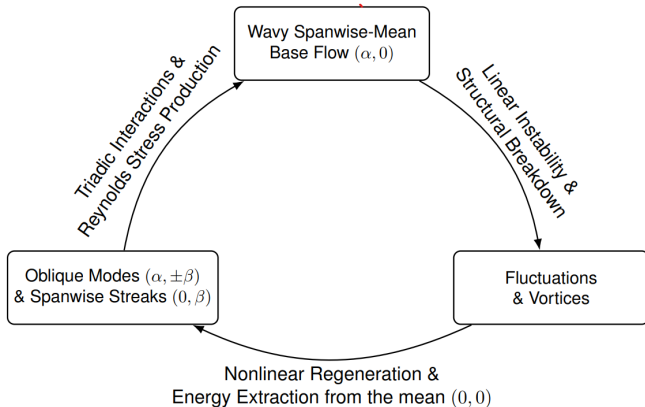


Energy evolution with the oblique modes removed (left) and with the streamwise streaks removed (right)
(streak energy (—), wavy spanwise-mean mode energy (— —) and the vortex energy
(. . . .))

The New Spanwise-Driven SSP

A Self-Contained Cycle

- Complete self-sustaining cycle that bypasses the classical streamwise pathway



The new spanwise-driven self-sustaining cycle

Conclusion & Future Work

- Wall-bounded turbulence can self-sustain through spanwise-driven SSP
- Bypasses the canonical streamwise pathway
- **Open questions:**
 - What mechanism allows the oblique modes to extract energy from the mean flow?
 - Does this pathway persist at higher Reynolds numbers?
 - What role does this spanwise-driven SSP play in the overall turbulence maintenance mechanism?

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Thank you

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